Band-1 Hadamard Order 9540 – Classical Paley-I (q=9539); IN SPK via Paley-I; Gap 2052 Still OPEN

H_9540 Paley-I PASS; Paley-II SKIP (q=4769=19*251 NOT prime / not prime power); Kronecker FAIL/SKIP (4*2385; 2385===1 mod 4 NOT Hadamard-admissible); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-22

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-09-22_Band1_Hadamard_9492_PaleyI`; `2026-09-22_Band1_Hadamard_9480_PaleyI`; `2026-09-22_Band1_Hadamard_9468_PaleyI`.
Lab. `/workspace/experiments/band1/hadamard_9540_paleyI/`.
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Honesty stamp (non-negotiable).
This memo claims **finite construction of order 9540 only (verified max|HHˆT-nI|=0), via classical Paley-I q=9539 (prime ===3 mod 4). 9540 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 9540: Paley-I **PASS** ($q = 9539$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL** ($q = 4769 = 19 \cdot 251$ NOT prime / not prime power); Kronecker **FAIL/SKIP** (note $9540 = 4 \cdot 2385$; $2385 \equiv 1 \pmod 4$ NOT Hadamard-admissible). **9540** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **9512+9492+9480+9468+9420+9404+9392+9372+9344+9324+9320+9312+9284+9240+9228+9204-** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

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Abstract

Finite construction of order **9540**.

- H_{9540} = Paley-I ($q = 9539$) – classical SPK
- Paley-II for 9540: **SKIP** ($q = 4769 = 19 \cdot 251$ NOT prime / not prime power)
- Kronecker for 9540: **FAIL/SKIP** ($4 \cdot 2385$; $2385 \equiv 1 \pmod 4$ NOT Hadamard-admissible)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

```
python3 verify_hadamard_9540.py
```

Results. overall_pass=True; units 119/119; row blocks 75/75 (block size 128); wall 705.5364s; constructed max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 9540 = q + 1$ with $q = 9539$ prime $\equiv 3 \pmod 4$ -> Paley-I eligible.
- Paley-II: $q = 4769 = 19 \cdot 251$ NOT prime -> FAIL/SKIP.
- Kronecker: $9540 = 4 \cdot 2385$; $2385 \equiv 1 \pmod 4$ NOT Hadamard-admissible – FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_9540 Paley-I	Paley-I q=9539	0	697.356
H_9540 Paley-II	SKIP (q=4769=19*251 NOT prime / not prime power)	–	–
H_9540 Kronecker	FAIL/SKIP (4*2385; 2385===1 mod 4 NOT Hadamard-admissible)	–	–

SPK closure note

Seeds include prior SPK through **9492** lineage (**9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; . . . ; **NOT** 8948/8644/8516/. . .) + verified **9540**. Next constructive gap **2052** (OPEN).

Quantity	Value
9540 in SPK	**True** (Paley-I q=9539)

Quantity	Value
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q=9547->n=9548 Paley-I DIRECT q=9547 OR q=9551->n=9552 Paley-I DIRECT q=9551 OR q=9587->n=9588 Paley-I DIRECT q=9587 OR n=2052 standard SPK constructive gap – method-attack / lit OR Track B / Object 1 meter await